

Chapter five

5. Computer virus

5.1. What is security?

Security is the protection of information from theft or corruption.

Types of Security

1. Network security
2. System and Software security and
3. Physical security

Achieve security: - A computer security is achieved by the following: -

- ✓ Cryptography (encryption/encode and decryption/decode)
- ✓ Secure network
- ✓ Anti-viruses and
- ✓ firewalls

5.2. Definition of Computer Virus

A **Computer Virus** is a malicious program that is installed into a computer without the knowledge of its user. This program can replicate itself and infect all the programs and files that are in the system. These viruses make the victim's computer malfunction or, at the worst, completely unusable.

A computer virus is a program which can harm our device and files and infect them for no further use. When a virus program is executed, it replicates itself by modifying other computer programs and instead enters its own coding. This code infects a file or program and if it spreads massively, it may ultimately result in crashing of the device.

Across the world, Computer viruses are a great issue of concern as they can cause billions of dollars' worth harm to the economy each year.

A computer virus is a piece of code embedded in a legitimate program and is created with the ability to self-replicate infecting other programs on a computer. Just like how humans catch a cold or flu, it can remain dormant inside the system and gets activated when you least expect it.

A computer virus is developed to spread from one host to another and there are numerous ways on how your computer catches it. It can be through email attachments, file downloads, software installations, or unsecured links.

These viruses can steal your data such as passwords, hacked into your social media accounts or online banking accounts, and even wiped out all your data.

5.3. Types of Computer Virus

Viruses can be classified according to the method that they use to infect a computer. Discussed below are the different types of computer viruses:

- 1 **Boot Sector Virus** – It is a type of virus that infects the boot sector of floppy disks or the Master Boot Record (MBR) of hard disks. The Boot sector comprises all the files which are required to start the Operating system of the computer. The virus either overwrites the existing program or copies itself to another part of the disk. These viruses are once common back when computers are booted from floppy disks. Today, these viruses are found distributed in forms of physical media such as external hard drives or USB. If the computer is infected with a boot sector virus, it automatically loads into the memory enabling control of your computer.
- 2 **Resident Virus** – A virus which saves itself in the memory of the computer and then infects other files and programs when its originating program is no longer working. This virus can easily infect other files because it is hidden in the memory and is hard to be removed from the system. A resident virus stores itself on your computer's memory which allows it to infect files on your computer. This virus can interfere with your operating system leading to file and program corruption.
- 3 **Multipartite Virus** – A virus which can attack both, the boot sector and the executable files of an already infected computer is called a multipartite virus. If a multipartite virus attacks your system, you are at risk of cyber threat. A type of virus that is very infectious and can easily spread on your computer system. It can infect multiple parts of a system including memory, files, and boot sector which makes it difficult to contain
- 4 **Overwrite Virus** – One of the most harmful viruses, the overwrite virus can completely remove the existing program and replace it with the malicious code by overwriting it. Gradually it can completely replace the host's programming code with the harmful code.
- 5 **Polymorphic Virus** – Spread through spam and infected websites, the polymorphic virus are file infectors which are complex and are tough to detect. They create a modified or morphed version of the existing program and infect the system and retain the original code. This virus has the capability to evade anti-virus programs since it can change codes every time an infected file is performed.

- 6 **File Infecting Virus** – As the name suggests, it first infects a single file and then later spreads itself to other executable files and programs. The main source of this virus is games and word processors. A virus that attached itself to an executable program. It is also called a parasitic virus which typically infects files with .exe or .com extensions. Some file infectors can overwrite host files and others can damage your hard drive's formatting.
- 7 **Spacefiller Virus** – It is a rare type of virus which fills in the empty spaces of a file with viruses. It is known as cavity virus. It will neither affect the size of the file nor can be detected easily.
- 8 **Macro Virus** – A virus written in the same macro language as used in the software program and infects the computer if a word processor file is opened. Mainly the source of such viruses is via emails. This type of virus is commonly found in programs such as Microsoft Word or Excel. These viruses are usually stored as part of a document and can spread when the files are transmitted to other computers, often through email attachments.

5.4. Symptoms that an Infected Computer Shows

Since the computer virus only hits the programming of the device, it is not visible. But there are certain indications which can help you analyze that a device is virus-hit. Given below are such signs which may help you identify computer viruses:

- a) **Speed of the System** – In case a virus is completely executed into your device, the time taken to open applications may become longer and the entire system processing may start working slowly
- b) **Pop-up Windows** – One may start getting too many pop up windows on their screen which may be virus affected and harm the device even more
- c) **Self-Execution of Programs** – Files or applications may start opening in the background of the system by themselves and you may not even know about them
- d) **Log out from Accounts** – In case of a virus attack, the probability of accounts getting hacked increase and password protected sites may also get hacked and you might get logged out from all of them
- e) **Crashing of the Device** – In most cases, if the virus spreads in maximum files and programs, there are chances that the entire device may crash and stop working

The first thing which you might notice in case of virus attack is the speed with which your system shall process. And then gradually other changes can also be observed.

5.5. Prevention Mechanism of Computer Virus

Just like how we protect ourselves from catching a virus, here are a few notes to remember to help keep your devices safe. There are several ways to protect your computer from viruses, including:

- 1 Avoid clicking on suspicious links.
- 2 Scan email attachments before opening it.
- 3 Avoid clicking on pop-up advertisements and get a pop-up blocker for your web browser.
- 4 When routed to another website, don't do anything, and immediately leave the site.
- 5 Install a reliable anti-virus program and always keep it up to date. The most suitable way of making your computer virus-free is by installing Anti-virus software.

Use a Trusted Antivirus Product

Trusted computer antivirus products are crucial to stop malware attacks and prevent computers from being infected with viruses. These antivirus concepts will protect devices from being infected through regular scans and identifying and blocking malware.

Avoid Clicking Pop-up Advertisements

Unwanted pop-up advertisements are more than likely to be linked to computer viruses and malware. Never click on pop-up advertisements because this can lead to inadvertently downloading viruses onto a computer.

Scan Your Email Attachments

A popular way to protect your device from computer viruses is to avoid suspicious email attachments, which are commonly used to spread malware. Computer antivirus solutions can be used to scan email attachments for potential viruses.

Scan the Files That You Download Using File-sharing Programs

File-sharing programs, particularly unofficial sites, are also popular resources for attackers to spread computer viruses. Avoid downloading applications, games, or software from unofficial sites, and always scan files that have been downloaded from any file-sharing program.

What is an Anti-Virus?

An anti-virus is software which comprises programs or set of programs which can detect and remove all the harmful and malicious software from your device. This anti-virus software is designed in a manner that they can search through the files in a computer and determine the files which are heavy or mildly infected by a virus.

Given below is a list of few of the major antivirus software which is most commonly used:

- ✚ Norton Antivirus
- ✚ F-Secure Antivirus
- ✚ Kaspersky Antivirus
- ✚ AVAST Antivirus
- ✚ Komodo Antivirus

These are few of the much anti-virus software widely used to remove viruses from a device.

5.6. Encryption

Data encryption is a way of translating data from plaintext (unencrypted) to ciphertext (encrypted). Users can access encrypted data with an encryption key and decrypted data with a decryption key.

- ✚ Encryption is a way to conceal information by altering it so that it appears to be random data. Encryption is essential for security on the Internet.
- ✚ Encryption is a way of scrambling data so that only authorized parties can understand the information.
- ✚ In technical terms, it is the process of converting human-readable plaintext to incomprehensible text, also known as ciphertext.
- ✚ In simpler terms, encryption takes readable data and alters it so that it appears random.
- ✚ Encryption requires the use of a cryptographic key: a set of mathematical values that both the sender and the recipient of an encrypted message agree on.
- ✚ Encryption is the basic building block of data security. It is the simplest and most important way to ensure a computer system's information can't be stolen and read by someone who wants to use it for malicious purposes.
- ✚ The more complex the cryptographic key, the more secure the encryption – because third parties are less likely to decrypt it

What are the most common techniques of encryption?

The two most common encryption methods are **symmetric** and **asymmetric** encryption. The names refer to whether or not the same key is used for encryption and decryption:

a) Symmetric encryption keys:

- ✓ This is also known as private key encryption.
- ✓ The key used to encode is the same as the one used to decode, making it best for individual users and closed systems. Otherwise, the key must be sent to the receiver.
- ✓ This increases the risk of compromise if it's intercepted by a third party, such as a hacker.
- ✓ This method is faster than the asymmetric method.

b) Asymmetric encryption keys:

- ✓ Also known as Public-Key Cryptography encrypts and decrypts the data using two separate cryptographic asymmetric keys. These two keys are known as a “public key” and a “private key”.
- ✓ This type uses two different keys — public and private — that are linked together mathematically.
- ✓ The keys are essentially large numbers that have been paired with each other but aren't identical, hence the term asymmetric.
- ✓ The private key is kept secret by the owner, and the public key is either shared amongst authorized recipients or made available to the public at large.

5.7. Backup

- ✚ Backup is **the process of creating a copy of the data on your system that you use for recovery in case your original data is lost or corrupted.**
- ✚ Storing the copy of the data on separate medium is critical to protect against primary data loss or corruption. This additional medium can be as simple as **an external drive or USB stick, or something more substantial, such as a disk storage system, cloud storage container, or tape drive**

Why people create computer viruses?

Unlike biological viruses (see Virus), computer viruses do not simply evolve by themselves. Computer viruses do not come into existence spontaneously, nor are they likely to be created by

bugs in regular programs. They are deliberately created by programmers, or by people who use virus creation software.

Computer viruses can only do what the programmers have programmed them to do.

Virus writers can have various reasons for creating and spreading malware. Viruses have been written as research projects, pranks, vandalism, to attack the products of specific companies, to distribute political messages, and financial gain from identity theft, spyware, and crypto viral extortion. Some virus writers consider their creations to be works of art, and see virus writing as a creative hobby.